LABSHEET 6

BACK-END INFRASTRUCTURE PREPARATION & SHELL SCRIPTING

Demonstrate installation and workspace environment path mappings for Python runtimes, Django frameworks, and Visual Studio Code IDE setups.

Task 6.1: Setup an isolated Django virtual execution ecosystem using venv clusters and pipe project requirement definitions into a formal setup configuration script.

Django Virtual Environment

Windows CMD/PowerShell:

python -m venv venv

venv\Scripts\activate

pip install django

pip freeze > requirements.txt

django-admin startproject myproject

cd myproject

python manage.py runserver

requirements.txt

Django

Task 6.2: Map custom user shortcuts and syntax automation workflows inside the VS Code development profile directory specifically for Python PEP8 compliance.

VS Code PEP8 Setup

Install the Python formatter:

pip install black

Create **.vscode/settings.json**:

```
{
    "editor.formatOnSave": true,
```

```
    "python.analysis.typeCheckingMode": "basic",
    "editor.defaultFormatter": "ms-python.black-formatter"
}
```

Task 6.3: Script an automated deployment check file that validates systemic runtime pathways, framework dependencies, and global server environment keys.

Automated Deployment Check

Create **check_env.py**:

```
import sys
import os
import django

print("=== DEPLOYMENT CHECK ===")

print("Python:", sys.version.split()[0])
print("Django:", django.get_version())

print("Python Path:", sys.executable)

if os.getenv("DJANGO_SETTINGS_MODULE"):
    print("DJANGO_SETTINGS_MODULE: OK")
else:
    print("DJANGO_SETTINGS_MODULE: Missing")
print("Deployment check completed.")
```

Output

```
=== DEPLOYMENT CHECK ===
Python: 3.x.x
Django: 5.x.x
Python Path: ...\venv\Scripts\python.exe
DJANGO_SETTINGS_MODULE: OK
Deployment check completed.
```